

Question 1 [10 Marks]

A software company is planning the development timeline for a cloud-based application. To estimate the duration of a critical module, the project manager collected the following estimates:

- If technical complications and integration issues occur, the task could take as long as 20 days.
- Under normal working conditions, the team expects the task to take around 9 days.
- If everything goes smoothly with no major interruptions, the task may finish in 5 days.

To obtain a more realistic schedule estimate, the company decides to use probabilistic duration estimation techniques.

Tasks:

- a) Using both Triangular Distribution and Beta Distribution (PERT), calculate the expected duration of the activity. Show all calculations clearly. [6]
- b) Explain why probabilistic estimation techniques are more suitable for software development projects than using a single fixed estimate. [4]

Question 2 [10 Marks]

A technology company is planning its operational workflow for launching a new AI-based service. To ensure smooth operations and timely service delivery, the operations team wants to identify the sequence of activities that may directly affect the overall completion time of the workflow. Therefore, the company decides to use the Critical Path Method (CPM) for schedule analysis. The operational activities are given below:

Activity	Duration (Days)	Immediate Predecessor
A	2	—
B	5	A
C	4	A
D	6	B
E	3	C
F	2	D, E

Tasks:

- Construct the logical sequence of the project activities using a network representation. [4]
- Determine the Critical Path using float and the minimum project completion time. [4]
- Briefly explain why delaying a critical activity affects the entire project schedule.[2]